

Teamwork Agreement

For the MathWorks Classroom Challenge Projects

Project: Data Analytics Apps for a Smart Eco Gym

Team Name: Team #2

Instructors: Kevin Graham, Dr. Harish Chintakunta, Dr. Esperanza Linares Guerrero

Program/Class: Engineering Pathway Program 2026

Members and Roles

1. Melvin Tam – Project Manager (PM)
2. Robert LeDoux – Modeling Lead
3. Allison Oh – Documentation & Visualization Lead
4. Miguel Peinado – Analysis/Validation & Quality Assurance

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Melvin- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Robert- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Allison- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Miguel- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Text message
- Secondary channel: Discord
- Response time: Depending on urgency, 2-5 hours on text and 6-12 on discord on weekdays and 24 hours on weekends.
- Meeting cadence: Tuesday and/or Thursday afternoons unless conflict occurs for 45 minutes on Zoom
- Decision records: See Team Decisions tab

4) Collaboration & Tools

- Version control: _____ Shared GitHub Repo_____
- Data management: All datasets, parameters, and assumptions documented in _____ README.md_____

File naming & style: Snake Case (e.g. Snake_case, descriptive names)

- Backups: ____tag Github versions____

5) Quality Standards (Definition of 'Done')

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1 (7/13-20): Review and understanding project and requirements, getting used to MATLAB and GitHub, mini challenge selected
 - Goal: have a dataset and understanding of it.
- Week 2 (7/20-27): cleaning data, getting ready to use in code for analysis. Deciding user features.
- Week 3 (7/27-8/3): coding analysis and frontend development. validation/quality assurance
- Week 4 (8/3-10): gathering report, making figures, slides, video and demo presentation

7) Decision-Making & Conflict Resolution

- Default: Consensus after discussion
- If blocked >48 hrs: majority vote, PM documents in Team Decisions tab

- Technical disagreements: Evidence-based comparison before vote
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: Master Google Document
- Accountability: Missed Deadline - Set a meeting time to discuss new plan
- Attendance: Let members know before the team meeting (6-8 hours beforehand) through text

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Code repository with clear structure and README ready for judging
- Code is representative of hand calculations and problem solving
- Final figures and report, ensuring all work is cohesive and relevant to our project
- Double checking project requirements in GitHub repository

12) Agreement & Signatures

Name & Signature: Allison Oh Date: July 14th, 2026

Name & Signature: Melvin Tam Date: July 14th, 2026

Name & Signature: Robert LeDoux Date: July 14th, 2026

Name & Signature: Miguel A. Peinado Date: July 14th, 2026